Sangmin Lee

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Education

M.S. **Korea Advanced Institute of Science and Technology (KAIST)**, Graduate School of Data Science (GSDS), KIRC Lab — Daejeon, Korea — Mar 2027 – Feb 2029
- Research interests: fine-tuning Large Language Models (LLMs) specialized for the medical domain

B.S. **Jeonbuk National University**, Statistics, College of Natural Sciences — Jeonju, Korea — Mar 2021 – Feb 2027
- Total GPA: 4.40 / 4.5, Major GPA: 4.47 / 4.5

Experience

Vision and Learning Lab (UVLL), UNIST, Undergraduate Researcher — Ulsan, Korea
Advised by Prof. Seungryul Baek — Jan 2026 – July 2026 — 7 months
- Focused on 3D vision, human mesh recovery (HMR), and Vision-Language-Action (VLA)
- Studied SMPL-based human mesh recovery and motion reconstruction
- Surveyed and presented recent research trends in egocentric vision and VLA at lab seminars

Multiscale Lab, Jeonbuk National University, Undergraduate Researcher — Jeonju, Korea
Advised by Prof. Guebin Choi — Jan 2025 – Dec 2025 — 1 year
- Focused on data-adaptive approaches in multiscale analysis
- Studied deep learning applications in NLP and computer vision
- Prepared a research poster and participated in an academic conference

Jeonbuk National University, Teaching Assistant, Statistical Thinking and Society — Jeonju, Korea — Mar 2025
- Prepared and reviewed instructional materials

LG Aimers (6th), Participant — Jeonju, Korea — Jan 2025 – Feb 2025 — 2 months
- Completed coursework on Artificial Intelligence and Linear Algebra
- Participated in a data competition using infertility-related medical data

JBNU Big Data Club (JBIG), Member — Jeonju, Korea — Mar 2024 – Aug 2024 — 6 months
- Participated weekly in Dacon and Kaggle competitions, followed by code presentations during club seminars

Projects

Medical Image Analysis — Sept 2025 – present
- Collaborating with a university hospital on deep learning-based medical image analysis
- Developing AI models for clinical data analysis and diagnostic support
- Working on improving the accuracy and efficiency of medical imaging workflows

Deep Neural Networks for Protein Function Prediction — June 2025 – July 2025
- Developed and benchmarked deep learning approaches for enzyme function prediction using the ESM-2 protein language model
- Implemented ESM-2 + Random Forest and ESM-2 + Multi-Layer Perceptron architectures to predict Enzyme Commission (EC) numbers
- Achieved the best performance with the MLP approach (Micro F1: 0.89, Macro F1: 0.77), improving over similarity-based methods
- Applied hyperparameter optimization and class imbalance handling to improve performance on rare enzyme classes

Deep Learning-based Neural Granger Causality for Solar Radiation Data Mar 2025 – June 2025

- Addressed the limitations of traditional VAR models in capturing nonlinear dependencies in time series data
- Implemented component-wise MLP (cMLP) and LSTM (cLSTM) models, assigning an independent network to each variable
- Analyzed Granger causal relationships between solar radiation in Jeonju and key meteorological variables
- Identified causality structures consistent with established relationships in solar meteorology

Honors and Awards

Academic Excellence Scholarship, Sancheong-gun Local Scholarship May 2026
Received in 2021.05, 2024.05, 2024.11, 2025.05, 2025.11, and 2026.05
Sancheong-gun

Challenge Award, J-Point Scholarship Program July 2025
JBNU SW-Center

First Prize, 2024 Big Data Camp - Prompt Engineering Competition Jan 2025
COSS

Yellow Belt, Keun-Saram (Great Person) Project Dec 2024
Jeonbuk National University

Grand Prize, 2024 Jeonbuk National University AI Competition June 2024
Jeonbuk National University

Presentations

- Poster: Deep Learning-based Neural Granger Causality for Solar Radiation Data, The 2025 Summer Conference of the Korean Statistical Society, Gyeongju, Korea (Jun. 2025)

Certificates

SQLD: SQL Developer Dec 2025

ADsP: Advanced Data Analytics Semi-Professional Nov 2025

Computer Specialist in Spreadsheet & Database Level-1 Feb 2022

Skills

Programming
Python, R, C, C++

APIs & Frameworks
PyTorch, Scikit-learn, Transformers, Hugging Face